

Scaling the L2:

EL mempool
short term future

Csaba Kiraly - SF Gsil

June 10th, 2025

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How blobs propagate in the mempool today?

Mempool transaction diffusion 101

- a. Push
 - i. Depends on **TX type** ($\neq 3$) and **size** (< 4096 bytes)
 - ii. Pushing to **sqrt**(peercount) peers selected randomly
 - iii. “**Stable**” **random** peer selection, based on **<tx signer address, self ID, and peer ID>**
- b. Notify and Pull
 - i. Notify **all** (other) peers about tx: **<txhash, type, size>**
 - ii. Pull **scheduling** policy

How blobs propagate in the mempool today?

Measuring blob diffusion from a single Gsil node

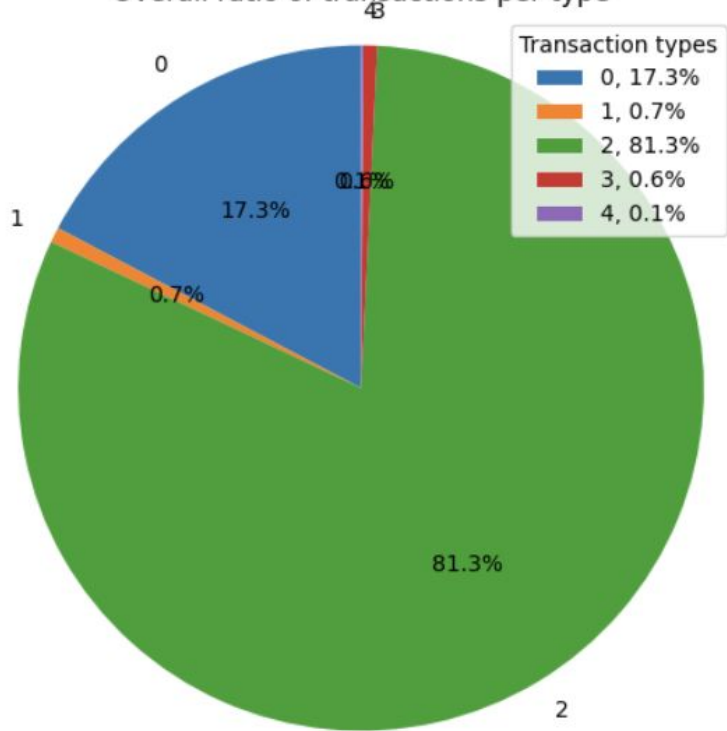
- For any given TX:

- a. Did we get it?
 - i. Push or pull, timestamped
- b. Did we see our peers having it?
 - i. Their announcement, receive timestamped
 - ii. From 50 peers from 500 peers
- c. Did we forward it?

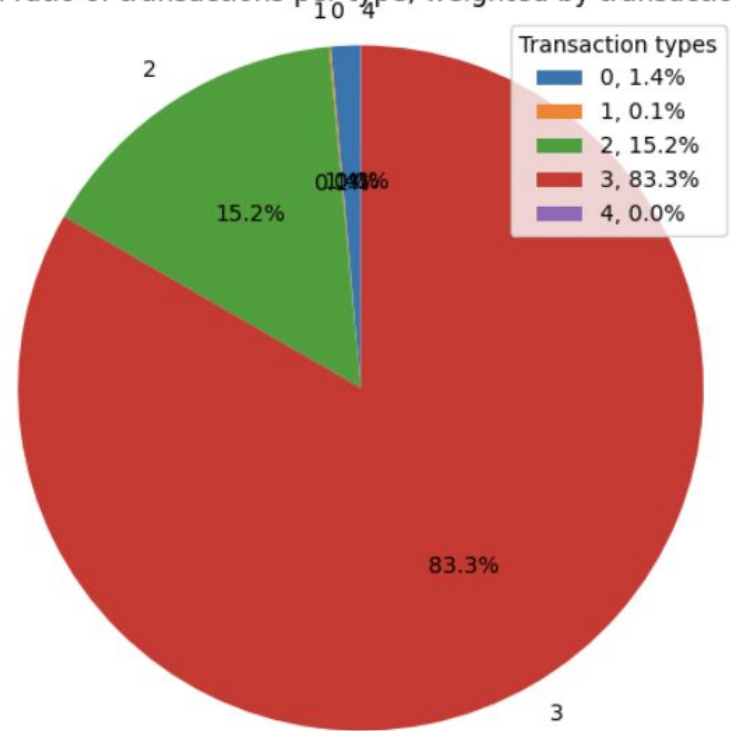
- When to evaluate?

Focus on “Block transactions”, evaluate when we get the block with the TX

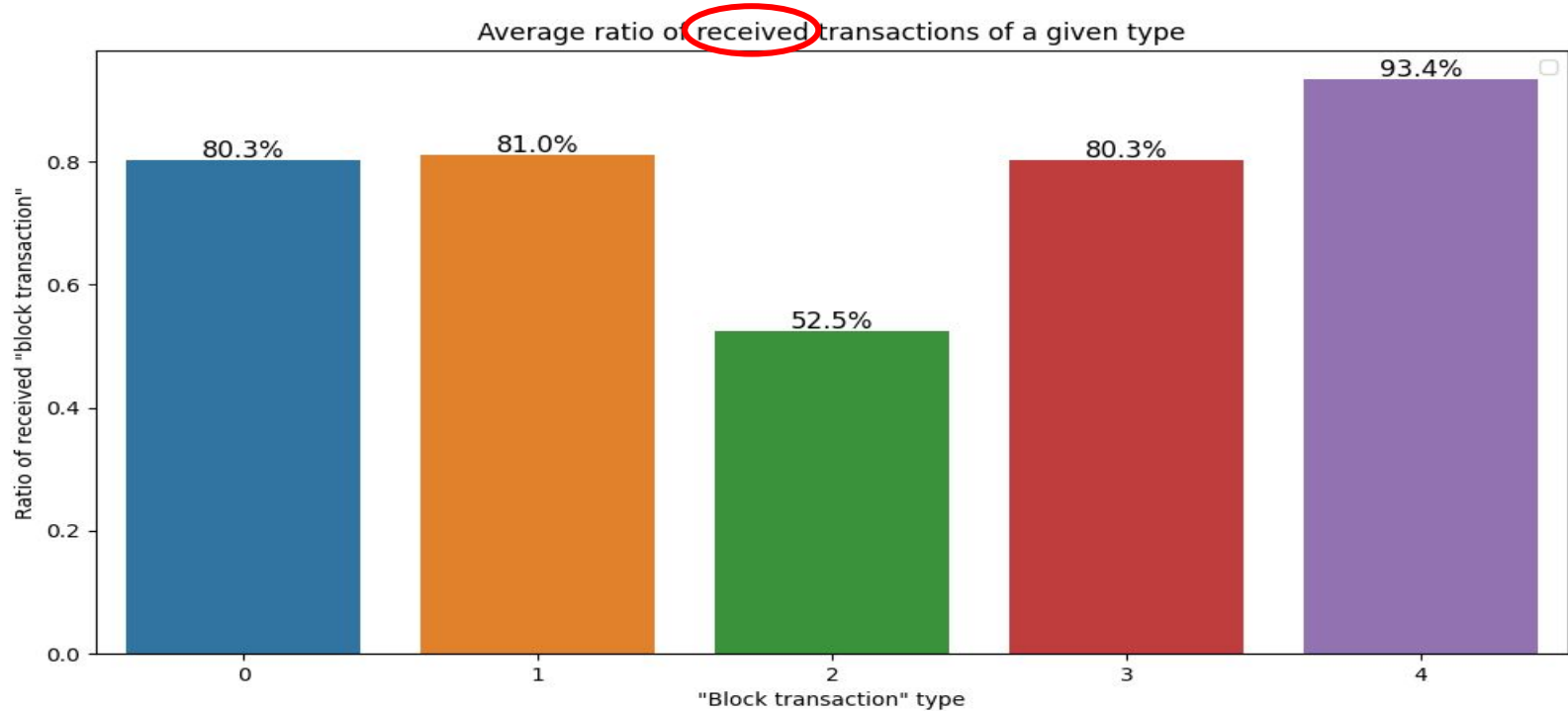
Overall ratio of transactions per type



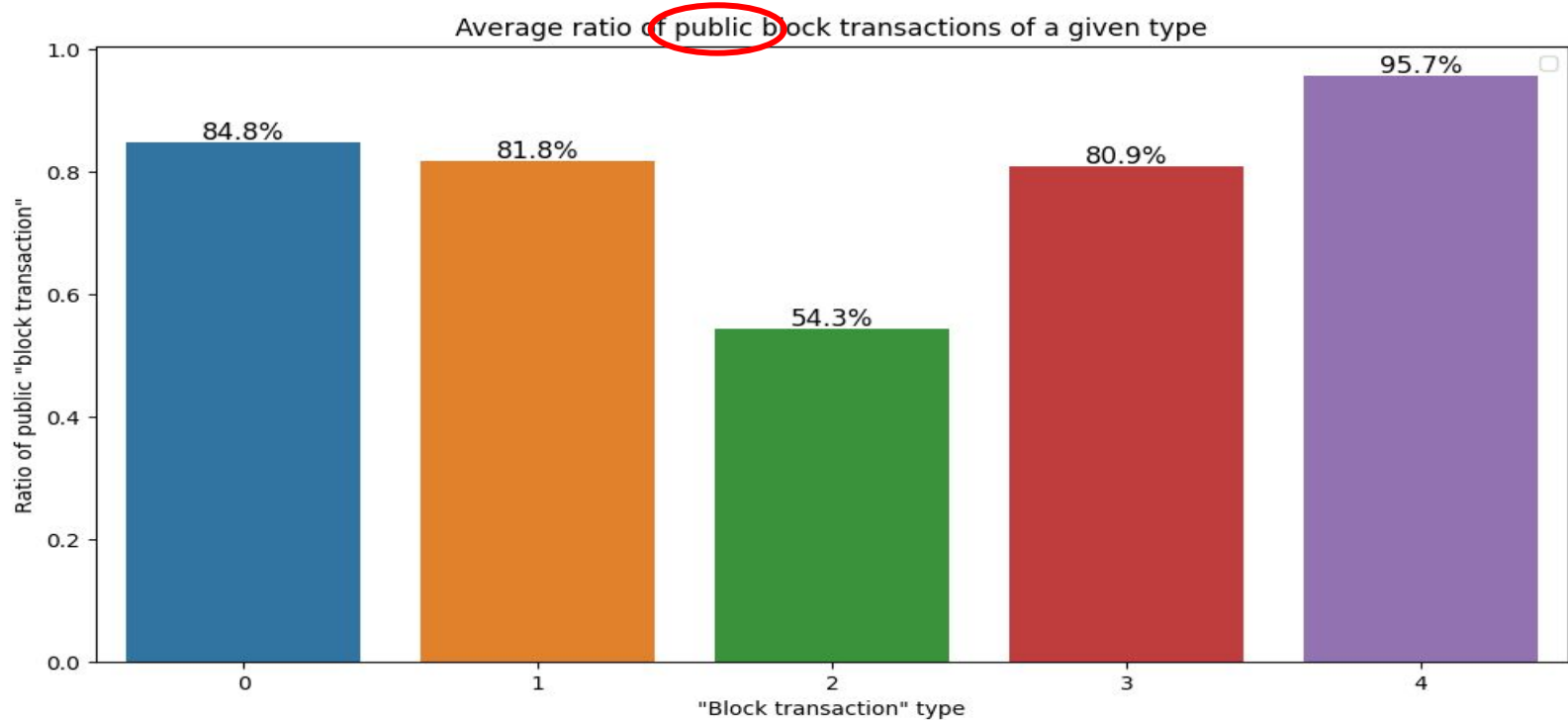
Overall ratio of transactions per type, weighted by transaction size



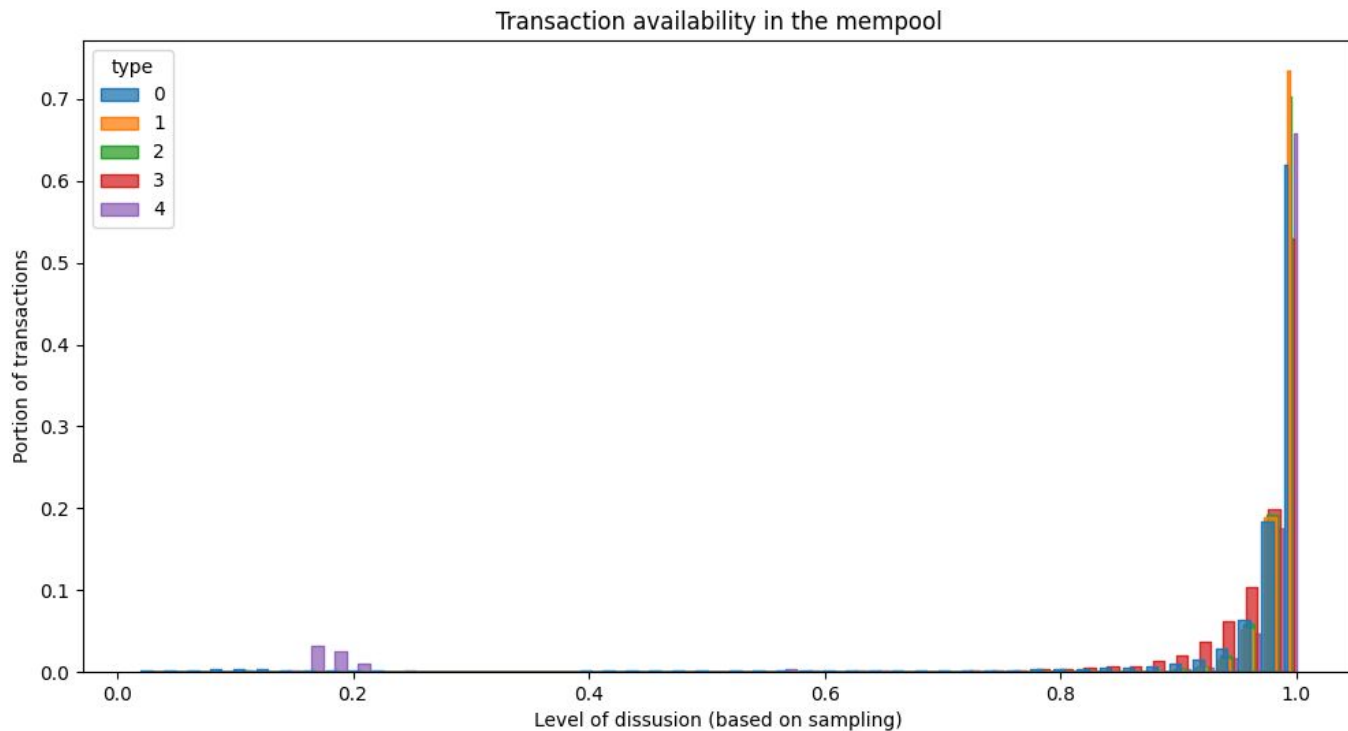
Received vs. Public



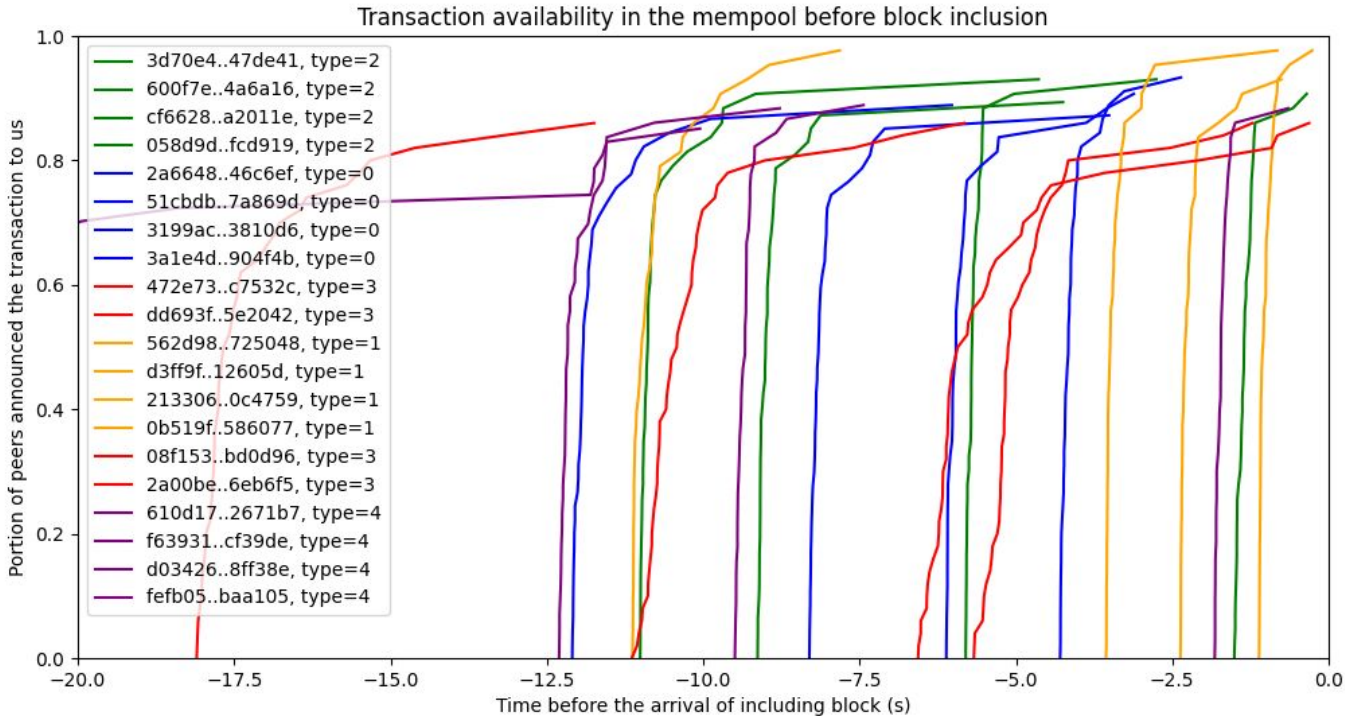
Received vs. Public



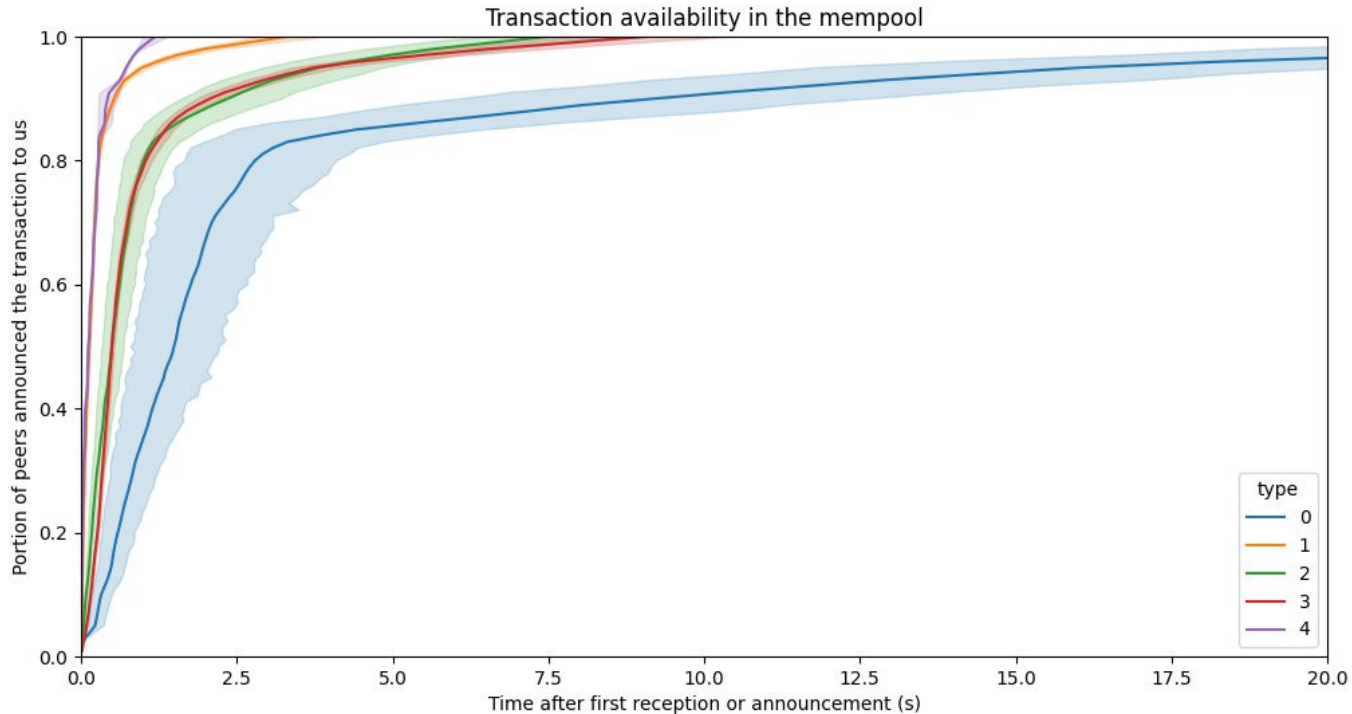
Diffusion ratio (0..1) of transactions in the mempool



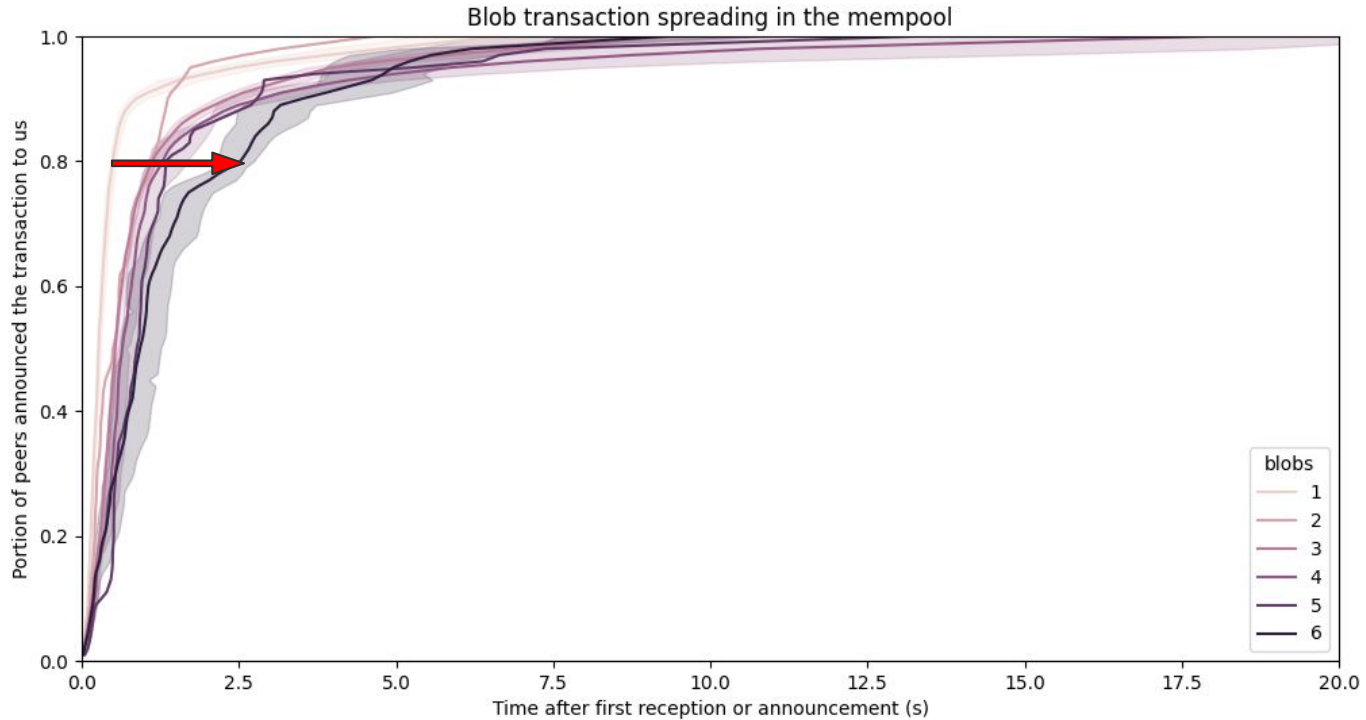
Transaction “diffusion speed” in the mempool



Transaction “diffusion speed” stats



Transaction “diffusion speed” stats

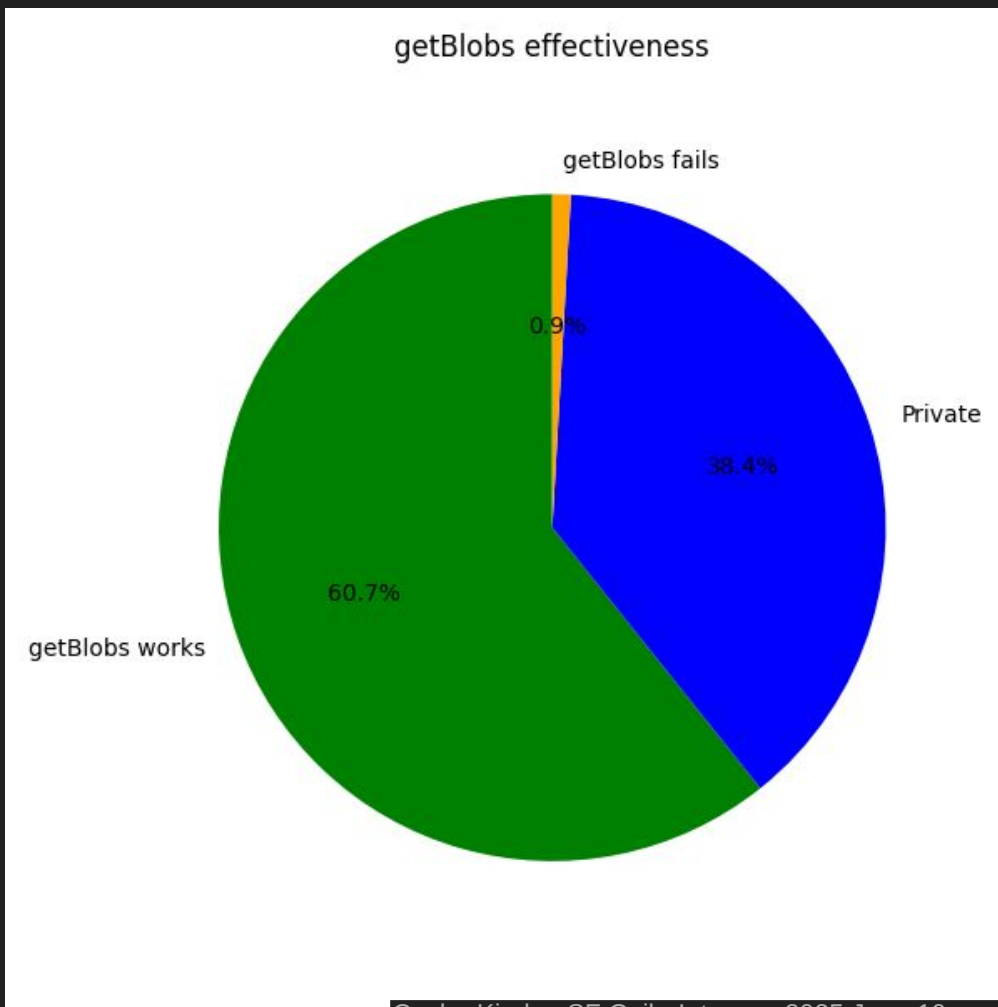


GetBlobs effectiveness

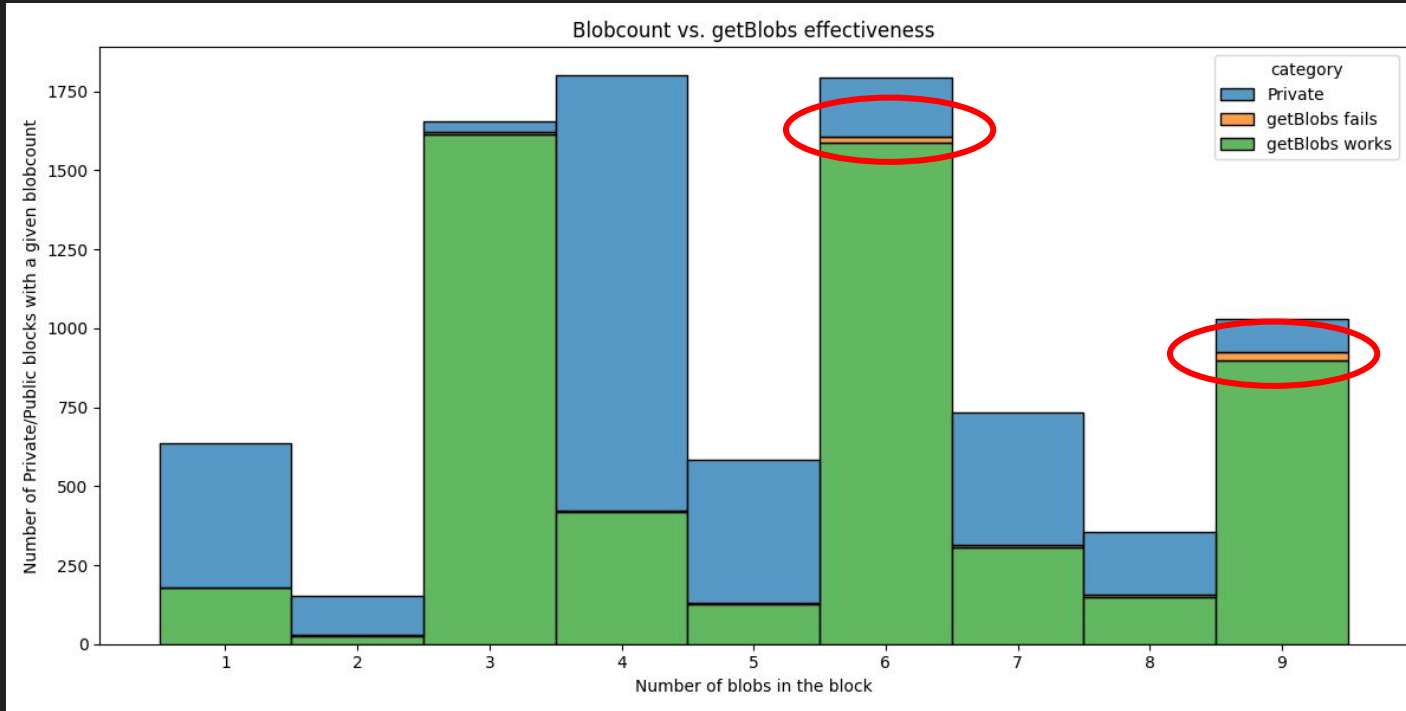
Private: it simply can't work,
choice of builder

Works: our node had all the blobs
by the time of block arrival

Fails: all blobs are public, but we
miss some



How would this scale with blob count?



Not enough info
here

But it feels OK

11 KB/sec/blob

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How blobs should propagate in the mempool **tomorrow?**

Fully synched vs. "sharded" (implicitly or explicitly)

Today

- the mempool feels basically **fully synched**

Tomorrow

- If we scale **WITHOUT changing** anything, it will be **implicitly sharded** (segmented)
 - Current policy is **TX sender address based sharding** (not yet applied to blobs)
 - We have **no practical experience** yet of how this would play out
 - We just know **it will happen** as an interaction of resource limits, announcement and pull scheduling policies

How blobs should propagate in the mempool **tomorrow?**

We can also introduce

- **Explicit horizontal sharding**, based on
 - TX hash
 - TX sender address
- **Vertical sharding**, based on blob segmentation and encoding
- **Sampling** in the mempool

Public mempool advantage (or what you miss with private blob feeds)

- pre-diffusion in the mempool (goes to DA and DAS through getBlobs today)
- sampling in the mempool?
- CR